Sigfox Access Station Micro
SMBS-T4

Coverage where you need it, made easy!

Improve Sigfox device coverage indoor and outdoor

Easy to install

Cost efficient

The **Sigfox Access Station Micro** provides affordable IoT service coverage to thousands of Sigfox devices. It targets operator densification and customer premises as an extension of the public network in areas such as deep indoor or remote locations.

Version : June 2019

Typical applications

- Customer specific coverage (Supply Chain and Logistics, Industry, Building Management, Retail, Agriculture, Mining…)
- Operator network densification
- Device and Solution development
- Demonstrations and Proof of concept

Key features

- All Radio Configurations in operation
- Mounting options: table, column/pole and wall
- Covers several floors up to entire building
- Management tools:
  - Locally with Android app **Access Station Utility**
  - Remotely via Sigfox OSS platform (configuration, monitoring, firmware upgrade)[1]

Highlights

Coverage … Where you need it

Discreet with integrated antenna and sleek design, Sigfox **Access Station Micro** can be installed at customer location: hidden or on display, indoor or rural outdoor, in an attic or underground, in warehouses, buildings, shops, farms or offices…

With its sealing cover, the **Access Station Micro** is dust and water resistant (IP65).

Coverage… Made easy

With its compact form factor and integrated antenna, **Sigfox Access Station Micro** can be easily installed within minutes. Accessories are included in the box to start using the Access Station Micro straight away.

Power-over-Ethernet allows flexible installation and do not necessarily require professional assistance.

Sigfox **Access Station Micro** connects to Sigfox Cloud via existing LAN infrastructure and Internet access but it also has the possibility to use cellular networks for quick and versatile installations.

Coverage … Affordable

Sigfox Access Station Micro addresses the challenges of affordable coverage densification by keeping the RoI of your IoT solution deployment and operations low.

Designed with great sensitivity it can cover an entire building[2] and can process up to 70 000 frames per day[3].

With extremely low energy consumption, the Access Station Micro is compatible with smart solar panel deployment enabling untapped application connectivity where no power source is available.

1. Subject to Operator control
2. Subject to building materials and any obstructions or interference present
3. Subject to available bandwidth and installation guidelines

Specifications

RADIO CHARACTERISTICS	
Standard	Sigfox Ultra Narrow Band Protocol for M2M and IoT
Max range of operating frequencies*	865 to 928 MHz
Max Receiver Sensitivity	-132dBm @ 100bps / -124dBm @ 600bps
Data Rate and Modulation	100 / 600 bps D-BPSK (UL), 600 bps GFSK (DL), OOK (Monarch beacon)
Max Transmit Power (EIRP)*	23 dBm ± 1dB
Antenna	Integrated
INTERFACES	
Ethernet	1 x RJ45 (10/100BaseT)
USB port	1 x USB 2.0 female type A
POWER	
Power Consumption	2.3W typical (Rx mode, Ethernet) / 7.5W max peak **
Power supply	Passive PoE with AC 220/110V adaptor in package
Input DC voltage	11 to 26V
MECHANICAL AND ENVIRONMENTAL	
Casing dimensions (W, H, D)	186 x 159 x 108 mm (7.3 x 6.3 x 4.25 in)
Product weight	450 g (1 lb)
Operating temperatures	-20°C to +55°C (-4°F to 131°F)
Storage temperatures	-30°C to +85°C (-22°F to 185°F)
Robustness	MTBF 92,000 hours
Protection	IP65 (with sealing cover)
Casing material	Plastic ASA/PC
COMPLIANCE	
Safety	EN 60950-1, IEC 60950-1 ; EN 62368-1, IEC 62368-1
Radio	EN 300 220-2 ; EN 300 220-1 ; FCC part 15.247 ; ARIB STD-T108
EMC	EN 301 489-3 ; EN 301 489-1 FCC Part 15 B ; FCC 15.207 and FCC 15.209

(*) The maximum frequency range and power setting will vary by channel and according to country regulations. Refer to the regulatory groups table for more details.
(**) Cellular dongle increases peak power consumption and is model dependant (5W max).

Regulatory groups are a simple way of grouping countries that have common policies related to this radio frequency equipment (power level, frequency band, access mode).
The regulatory groups used by Sigfox are simply referred to by a single letter.
To identify the regulatory group that corresponds to a particular country please consult the table below.
Not all regulatory domains / countries may have been approved for all Sigfox products. Please check the status with your local sales representative.

VARIANT	REGULATORY GROUP	RADIO ACCESS MODE	OPERATING BAND	MAX OUTPUT RADIATED POWER (EIRP)	COUNTRY
SMBS-T4	A	FH	915 - 928MHz	23 dBm	AUSTRALIA
	B	FH	902-907.5 & 915-928 MHz	23 dBm	BRAZIL
	C	FH	915 - 928MHz	23 dBm	CHILE COLOMBIA URUGUAY
	E	DC 10%	869.4-869.65MHz	23 dBm	EUROPE (EU) KENYA MAURITIUS OMAN SOUTH AFRICA SWAZILAND TUNISIA TURKEY UAE
	H	FH	920 - 925MHz	23 dBm	HONG KONG
	I	DC 10%	865 - 867 MHz	23 dBm	INDIA
	J	LBT	920.6-922.2MHz	23 dBm	JAPAN
	K	LBT	922.1 - 923.4MHz	14 dBm	SOUTH KOREA
	M	FH	919 - 923 MHz	23 dBm	MALAYSIA
	N	FH	920-928MHz	23 dBm	NEW ZEALAND
	P	FH	916-928MHz	23 dBm	PERU
	R	FH	920.5 - 928 HHz	23 dBm	COSTA RICA
	S	FH	920-925MHz	23 dBm	SINGAPORE
	U	FH	902-928MHz	23 dBm	ARGENTINA CANADA HONDURAS MEXICO PANAMA PUERTO RICO USA
	T	FH	920-925MHz	23 dBm	TAIWAN

DC: Duty Cycle - FH: Frequency Hopping - LBT: Listen Before Talk

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